

Bootstrapping the Zero-Coupon Yield Curve for G-Secs

A cubic-spline ZCYC bootstrap, a comparison against FBIL's published curve,
and bonus SDL STRIPS pricing.

Abinav Thangaraju Sethupathy

Khaja Aflal

AS Kretik

Rishe Raghavendra Gnanasekaran

01

Data Extraction & Filtering

Sourcing, cleaning & node selection

What FBIL Actually Publishes

01 · DATA

FBIL GOI Prices/Yields · 11-Sep-2026 · sheet “G-Sec” · 124 ISINs. One row, as published:

ISIN	Coupon	Maturity	Price	YTM (semi)
IN0020230119	7.33%	30-Oct-2026	100.2771	5.009%

Two remark columns in the same file shape everything that follows:

- **5 rows are floating-rate bonds (FRB).** A floater’s quoted YTM is not a fixed-coupon yield, so it cannot sit on a nominal zero curve. These are dropped first.
- **26 rows are flagged “Input Point”.** This is FBIL’s own node set for the day. We do not use it for our curve, but it gives us a ready-made benchmark to test our own selection against.

The zero and par halves of FBIL’s curve come from two separate workbooks, joined on tenor, and T-bill rates from a third feed. All are public and for the same date.

How FBIL Decides What a Yield “Is”

01 · DATA

Level	Qualifying evidence	What it contributes
1	At least 3 NDS-OM trades (1–14y) or 2 (beyond 14y) by 5 pm; yields beyond $\pm 2\sigma$ trimmed	Volume-weighted average yield (VWAY) of the surviving trades
2	Bid/offer quotes of at least ₹10 cr face, sampled 12/2/4 pm, spread within 5 bp	The volume-weighted bid–offer yield at 5 pm
3	Neither of the above clears the bar	The previous day’s proxy yield, carried forward

Model YTM + adjustment factor. When a bond clears neither Level 1 nor Level 2, FBIL does not publish a market yield for it. It publishes its own model yield plus an adjustment factor, roughly a 20-day moving-average illiquidity spread. So for most bonds, the published number is already partly the output of FBIL’s curve.

What We Replicate From the Concept Note

01 • DATA

- **The same three maturity segments**, split at 1 year and 14 years (§V). Bonds with 1 year or less to run are excluded, and the T-bills cover that stretch instead.
- **The same 90-day clustering rule** in the 1–14 year band (§VI): of any bonds maturing within 90 days of each other, only one survives. FBIL keeps the one with the highest volume $\times$ trades. The public file carries neither figure, so ours keeps the earliest-maturing bond.
- **The same six long-end buckets** beyond 14 years (14–18, 18–22, ..., 34+), one bond each, plus the longest-dated bond to anchor the far end (§VII).
- **The same short-end anchors**: 7-day, 6-month and 12-month T-bill rates, used as bond-equivalent yields.

Everything upstream of the published yield needs trade-level NDS-OM data, which is not public:

- the Level 1/2/3 qualification ladder,
- outlier trimming and the VWAY calculation,
- the adjustment factor for bonds that did not trade.

So we take FBIL's *published* YTM for every bond as given.

Partial circularity. An unlabelled column in FBIL's file marks only 25 of the 124 yields as trade-derived ("T") and 7 as proxies; the other 92 carry no mark. For a bond that did not trade, the yield we feed in is FBIL's model yield plus its adjustment factor. Comparing our curve with FBIL's is therefore partly comparing it with a curve built from its own output. Only the traded inputs are fully independent.

From 124 Rows to 40 Curve Inputs

01 · DATA

Valuation date. 11-Sep-2026 (Friday), settling T+1 on 14-Sep-2026. It was the newest date on which all five FBIL publications were in the free public archive.

Selection. 90-day thinning in the middle segment, one bond per long-end bucket, and three T-bills at the short end.

Segment	Universe	Selected
$\leq 1y$ (T-bills)	–	3
1–14y (middle)	76	30
$> 14y$ (long)	35	7
Total	119	40

124 published rows $\rightarrow$ 119 after dropping the 5 FRBs $\rightarrow$ 40 nodes for the cubic-spline bootstrap.

02

ZCYC Cubic Spline Approach

One smooth curve that reprices all 40 inputs

The Cubic Spline Bootstrap

The curve. The 40 input maturities are the knots $t_1 < \dots < t_n$; the unknowns are the zero rates z_i at those knots. Between two knots the curve is a cubic:

$$S_i(t) = a_i + b_i(t - t_i) + c_i(t - t_i)^2 + d_i(t - t_i)^3$$

Hits every knot	$S_i(t_i) = z_i, S_i(t_{i+1}) = z_{i+1}$	$2(n-1)$
Smooth slope	$S'_i(t_{i+1}) = S'_{i+1}(t_{i+1})$	$n-2$
Smooth curvature	$S''_i(t_{i+1}) = S''_{i+1}(t_{i+1})$	$n-2$
Natural ends	$S''(t_1) = S''(t_n) = 0$	2
Equations, matching the $4(n-1)$ coefficients		$4(n-1)$

From curve to prices.

$$DF(t) = (1 + z(t)/2)^{-2t}$$

$$P_j(z) = \sum_k CF_{jk} DF(t_{jk})$$

The market price is FBIL's clean price plus 30/360 accrued interest.

Solve. Choose all 40 knot rates together to minimise $\sum_j (P_j(z) - P_j^{\text{mkt}})^2$. Forty prices, forty unknowns: the fit is exact.

When it works. Knots distinct and increasing, one instrument per knot, at least 2 knots. Knots packed too close make the spline swing, hence the 90-day spacing.

Conventions Locked Before Fitting

02 · SPLINE

Decision	Choice
Settlement	T+1 (14-Sep-2026), rolled over weekends
Coupons	Semiannual, schedule counted back from maturity
Accrued interest	30/360 bond basis
Compounding	Semiannual; annualised rate = $(1 + z/2)^2 - 1$
Time to cash flow	Actual days / 365 from settlement
Short end	7D / 6M / 12M T-bills at 4.72% / 5.59% / 5.89%, each priced as a zero-coupon instrument
Output grid	0.25 to 50 years in quarter-year steps (200 points), matching FBIL's published grid
Par yield	$\text{par}(T) = 2(1 - DF(T)) / \sum_k DF(t_k)$

Honesty rule. All of this was fixed before any comparison with FBIL was run. The 50-year grid is not a deviation: \$X.5 asks for 40 years “or equivalent to the highest issued G-Sec”.

Spline Design Choices and Why

02 · SPLINE

Choice	Headline	Also tested	Why the headline
What we spline	Zero rates	Log discount factors	Directly comparable with FBIL's published zero curve
End condition	Natural	Not-a-knot	Keeps the curve from bending at the ends, where data is thinnest
Solver	Global, all at once	Sequential bootstrap	A coupon bond's cash flows fall between knots, so its price depends on the whole curve
Beyond the data	Flat	Cubic	A cubic extrapolation swings wildly with nothing to anchor it
Time measure	Actual/365	Whole half-years	Uses the real calendar

Checked on every run. The run stops if any input reprices worse than 10^{-6} per ₹100 (achieved: 1.6×10^{-13}), if curvature jumps at a knot, or if discount factors stop falling.

- **An approximation, not a replica.** FBIL's exact Pienaar–Choudhry formulation is not public. Ours is a cubic spline from the same family.
- **No holiday calendar.** Settlement rolls over weekends only.
- **The first knot sits at 7 days.** The Concept Note calls it both “overnight” and “7-day T-bill”. We use $7/365$ years and keep it as a setting.
- **T-bill tenors are nominal.** If FBIL quotes a 182- or 364-day bill, its rate is placed at exactly 0.5 or 1.0 years, a mis-timing of one day at most.
- **The fallback solver was not needed.** The sequential bootstrap only reprices to about 10^{-3} . On this date the global solve converged.

Long end. The longest G-Sec, 7.43% GS 2076, matures 49.35 years out. The last three grid points (49.5, 49.75 and 50 years) are held flat at its rate rather than fitted.

03

Testing

226 automated tests, all passing

What We Test, and Why

Stage	Tests	What passing proves
Bond maths	62	Coupon dates, 30/360 accrued interest and T+1 settlement are right, and price $\leftrightarrow$ yield converts both ways without loss
Node selection	36	The 90-day spacing, the 1-year boundary and the long-end buckets behave as the Concept Note says
Curve engine	59	Every input reprices, the curve is smooth, discount factors fall, and the output grid matches FBIL's format
Data loader	43	FBIL's files match our schema, and bad input stops the run instead of being quietly fixed
Comparison	22	Basis-point arithmetic, the sign convention (ours minus FBIL) and segment splits are right
STRIPS	4	RBI's worked examples are reproduced; STRIPS add up to min(book, market)

Separately, every curve checks itself inside the pipeline; one that fails never reaches the results.

1. **The engine recovers a curve we already know.** We priced 19 bonds off an invented “true” curve and gave the engine only the prices. It rebuilt the whole 0.25–50 year curve to within **0.15 bp**.
2. **Our bond maths matches FBIL’s prices.** Priced from FBIL’s own yields, all 119 bonds land a median **₹0.02** per ₹100 from FBIL’s price (worst ₹0.06): day-count rounding.
3. **We reproduce the regulators’ own examples:** the Concept Note’s 11-bond selection example (same 5 nodes) and RBI’s Annex 3 and 4 STRIPS examples (Annex 4: 127.87 normalised to 120.00).
4. **Two solvers, one curve.** Global and sequential solves agree; the global one is more accurate.

Clean data vs. real data. All 16 ablation variants build on the synthetic curve, but only 4 of 16 pass the built-in checks on FBIL’s real data: every run without 90-day thinning and every not-a-knot run fails. On real prices, spacing and natural ends are what keep the spline stable.

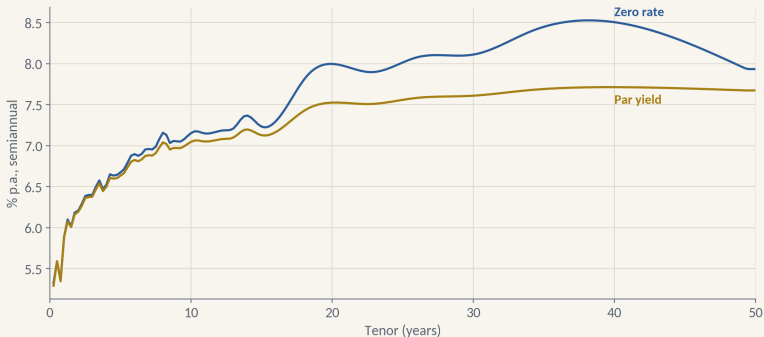
04

Results

Our curve against FBIL's, segment by segment

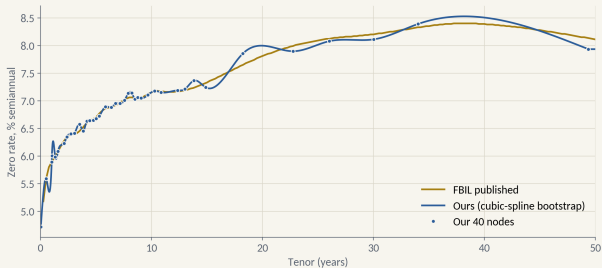
Our Zero-Coupon Curve, 14-Sep-2026

04 • RESULTS



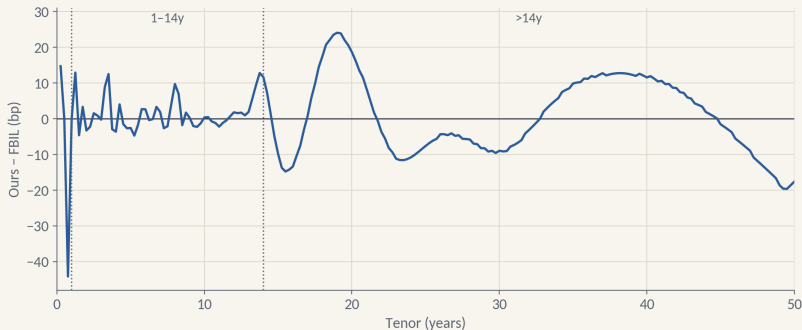
All 40 inputs reprice to within 1.6×10^{-13} per ₹100. Zero rates rise from 4.72% at 7 days to about 8.5% near 38 years. On an upward-sloping curve the par yield sits below the zero rate, as it does here at longer tenors.

Ours vs FBIL at a Glance



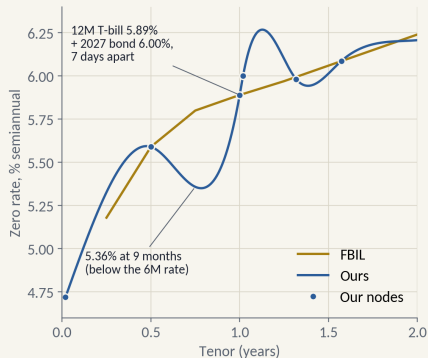
Across 200 grid points	≤1y	1–14y	>14y	Overall
RMSE (bp)	23.3	4.8	10.7	10.0
Mean absolute difference (bp)	14.7	3.4	9.3	7.9
Largest gap (bp)	44.2	12.9	24.1	44.2

Where the Gap Sits



Positive means our curve is above FBIL's. There is one sharp spike just under a year, small ripples through the middle, and slow waves beyond 14 years. Each has a different cause.

Up to 1 Year: One Misplaced Knot

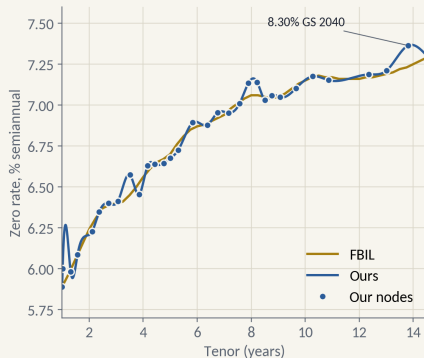


What we see. We match FBIL exactly at 6 and 12 months, where both curves use the same T-bills. In between, ours dips to 5.36% at 9 months, below the 6-month rate: a 44 bp miss.

Why. The 12-month T-bill (5.89%) and the 8.28% GS 2027 (6.00%) are knots only 7 days apart. To climb 11 bp in a week and stay smooth, the spline swings down before and up after. Our 90-day rule spaces bonds from each other, but not from the last T-bill.

Evidence. Four bonds maturing in 8–11 months miss our curve by 38–54 bp and FBIL's by 3–11 bp. Without that one bond, the gap falls from 23.3 to **0.5 bp**.

1 to 14 Years: Different Representatives



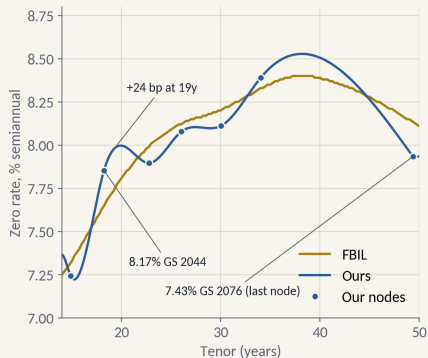
What we see. Our closest segment: 4.8 bp RMSE. The ripples are where our curve passes through a bond sitting slightly off its neighbours.

Why. FBIL keeps the most-traded bond in each 90-day cluster; we keep the earliest. At 13.8 years that gave us the 8.30% GS 2040, which FBIL's own curve misses by 7 bp; FBIL uses the 6.68% GS 2040 instead. And our spline hits every knot exactly, while FBIL's smooths past its inputs by about 4 bp.

Evidence. Rebuilt on FBIL's 26 Input Points, the gap falls from 4.8 to **1.9 bp**. On the 46 bonds neither curve uses as a knot, the two are within 1 bp of each other (6.5 vs 5.5 bp).

Beyond 14 Years: Few Bonds, Many Curves

04 • RESULTS



What we see. Two slow waves: 24 bp above FBIL at 19 years, 20 bp below at 49.

Why. Only 7 knots cover 35 years. Rates rise 61 bp between the 14.9- and 18.2-year knots, then flatten, and a natural spline overshoots that turn. The 18.2-year knot (8.17% GS 2044) is a bond FBIL's own curve misses by 9 bp. From 34 to 49 years there are no knots at all.

Evidence. FBIL's Input Points barely help (10.7 → 10.2 bp), so the cause is exact vs. smoothed fitting. And a 10–24 bp gap in zero rates is only 4.3 vs 2.1 bp on the 28 long bonds that are not knots.

What Explains the Gap

Segment	Cause	Checked by	RMSE, as built → test
≤1y	12-month T-bill and first bond knot 7 days apart	Dropping that one bond	23.3 → 0.5 bp
1–14y	Different representative bonds (no liquidity data)	Using FBIL's Input Points	4.8 → 1.9 bp
>14y	Sparse knots; exact spline vs. FBIL's smoothing	Input Points don't fix it	10.7 → 10.2 bp

Engine performance. Our curve reprices its 40 inputs exactly; FBIL's misses the same bonds by about 4 bp. On the 74 bonds with more than a year to run that neither curve uses as a knot, FBIL is slightly closer (4.5 vs 5.8 bp RMSE). That is expected: most of those yields are FBIL's own model output.

Diagnostics only. Both rebuilds test a cause; the headline curve stays exactly as locked before the comparison.

05

Comparison Analysis

Where our method departs from the source documents

Every Divergence, and Where It Stands

05 · DIVERGENCES

Concept Note rule vs. us	Status	Consequence
Liquidity decides which bond survives a 90-day cluster	Rule coded; data absent	Earliest maturity wins instead
Bonds with exactly 1 year left are excluded, not included	Fixed	Boundary tested
Node set chosen weekly (Monday), held for the week	Blocked	None for a single date
Level 1/2/3 trade qualification, VWAY, outlier trimming	Blocked	Every published bond is eligible
Published yields of untraded bonds are FBIL model output	Open	Some agreement is built in
Security prices carry an adjustment factor	Open	We compare curves, not bond prices
STRIPS rules written for Gol bonds on 2 Jan / 2 Jul	Open	Applied to an SDL by analogy (\$06)

Fixed: code now follows the document. Blocked: cannot be followed with public data. Open: a real tension, stated rather than resolved.

- **No liquidity data** → **off-curve knots**. Our tie-break keeps the earliest bond in each cluster or bucket. Three of our knots, the 8.28% GS 2027, 8.30% GS 2040 and 8.17% GS 2044, are bonds FBIL's own curve misses by 7–10 bp, and they sit behind the three largest bumps in our curve. Only 11 of our 37 bond knots are FBIL Input Points.
- **No trade qualification** → **circular inputs**. Because untraded bonds carry FBIL's model yield, FBIL's curve fits them better almost by construction. That is why FBIL wins the out-of-sample comparison by 1–2 bp.
- **No adjustment factor** → **exact fitting**. FBIL can let a single bond sit off its curve and absorb the difference in that bond's spread. Our spline must pass through every knot, so one odd bond bends the curve locally.

Is the first knot overnight or 7 days? The Concept Note calls the first short-end point “overnight” and then defines it as the 7-day T-bill rate published by FBIL. We place it at 7/365 years. At these tenors the difference is small, and it is a single setting.

Does the 90-day rule chain? If bonds A and B are within 90 days, and B and C are too, but A and C are 120 days apart, strict chaining drops C. The document’s own example never tests this. We keep both A and C, because the Concept Note’s stated aim is “minimum loss of data”, and every pair of knots is still more than 90 days apart.

Verified. On the Concept Note’s own worked example (11 bonds, dated 20-Nov-2019), our selection keeps the same 5 knots in the same clusters.

06

STRIPS Pricing (Bonus)

Splitting a State Development Loan into
zero-coupon pieces

The Task and the Bond We Chose

06 • STRIPS

The task. Pick a long-dated SDL and price each of its STRIPS using RBI's guidelines on stripping government securities.

The constraint. FBIL's published SDL zero curve stops at 14 years. Of the 5,438 SDLs in FBIL's file, 1,138 mature later than that, out to a Kerala 2064, and none of those can be discounted off a published curve.

Our rule. Take the longest SDL with between 1 and 14 years to run.

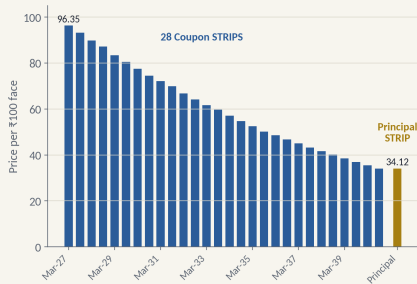
Security	7.47% Haryana SGS 2040
ISIN	IN1620250143
Maturity	10-Sep-2040, exactly 14.00 years (5,110 days)
Coupons	Semiannual, 10 Mar / 10 Sep
FBIL price	₹97.5774 clean, YTM 7.7566%
Accrued	4 days, ₹0.0830 → ₹97.6604 dirty
Stripped	₹5 crore face

1. **Split the bond.** Each remaining coupon becomes a Coupon STRIP and the redemption becomes a Principal STRIP: 28 coupons of ₹18,67,500 ($₹5 \text{ cr} \times 7.47\% \div 2$) and one principal of ₹5,00,00,000.
2. **Find each STRIP's zero rate** on FBIL's published SDL curve, spline-interpolated between its quarterly points. The March 2027 coupon, 0.485 years out, gets 5.80%.
3. **Discount it:** $PV_i = CF_i (1 + z_i/2)^{-2t_i}$.
4. **Normalise** (RBI ¶15.2): scale every PV by factor = $\min(\text{book}, \text{market}) / \sum_i PV_i$.
5. **Quote per ₹100 face**, the STRIP denomination (¶5).

Why normalise. At curve rates the STRIPS add up to ₹4,92,96,005.52, or ₹98.59 per ₹100. FBIL's market price is ₹97.66. The curve discounts early coupons at lower short-tenor rates (from 5.8% at six months), while the market price uses one 7.76% yield for every cash flow. Scaling every STRIP by the same factor makes the pieces add back to exactly what the bond is worth.

Results: Scenario 1, Book Value = Face

06 • STRIPS



Book value (face, ₹5 cr) exceeds market value (₹4,88,30,200), so **market value binds**: factor **0.99055**. Every input is FBIL data.

STRIP	Value (₹)	Per 100
Coupon Mar-27	17,99,245.41	96.35
Coupon Sep-27	17,41,671.06	93.26
Coupon Sep-33	11,14,875.11	59.70
Coupon Mar-40	6,62,079.20	35.45
Coupon Sep-40	6,37,151.98	34.12
Principal Sep-40	1,70,58,955.29	34.12
All 29	4,88,30,200.00	

The last coupon and the principal share a date, so they get the same price per ₹100. Coupons carry 65% of the bond's value.

Scenario 2: When Book Value Binds

06 · STRIPS

	Scenario 1	Scenario 2
Book value	₹5,00,00,000 (face)	₹4,73,65,294
Market value	₹4,88,30,200	₹4,88,30,200
Lower of the two	Market	Book
Normalisation factor	0.99055	0.96083
Mar-2027 coupon, per ₹100	96.35	93.45
Principal, per ₹100	34.12	33.09
STRIPS add up to	₹4,88,30,200.00	₹4,73,65,294.00

An illustration, not a result. Book value is the holder's own accounting figure and nobody publishes it. Scenario 2 sets it at $0.97 \times$ market value, the only assumed number in the whole project. It exists only to show the other branch of RBI's rule.

- **SDLs are not covered.** The guidelines apply to Government of India securities paying on 2 Jan / 2 Jul (¶10). Ours is state-issued and pays on 10 Mar / 10 Sep. The brief asks for something the guidelines don't strictly allow, so we apply them by analogy, using the bond's real coupon dates.
- **"Long-dated" vs. 14 years.** A 25-year SDL would need FBIL's curve extended past its published end, which means inventing the very rates the exercise is meant to source.
- **The market price is a tenor-level yield.** Every SDL maturing between July and December 2040 in FBIL's file carries the same 7.7566% yield, so the value we normalise to is not set by trades in this bond.
- **Out of scope:** HTM/AFS/HFT accounting (¶15.3–15.4), STRIPS ISIN generation, and reconstitution.

Verified. The engine reproduces RBI's Annex 3 and 4 examples; both scenarios add up to $\min(\text{book}, \text{market})$ to within 10^{-9} .

- A cubic-spline bootstrap that reprices all 40 inputs exactly, built from public FBIL data with the Concept Note's selection rules.
- **Within 10 bp of FBIL overall, and 4.8 bp between 1 and 14 years**, where most of the market trades.
- Every remaining gap traces to a cause we can test: one knot 7 days from a T-bill, no public liquidity data, and exact rather than smoothed fitting at the long end.
- A full STRIPS valuation of a 14-year SDL that adds up exactly to its market value, with every number from FBIL data.

Next step. Extending the 90-day rule to the T-bill boundary removes the short-end spike (23.3 → 0.5 bp). It is a one-line change we have not made to the locked headline.

Thank You